1) a)

$\det \mathbf{M} = \begin{vmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{vmatrix} = \cosh^2 x - \sinh^2 x = 1$

As $\det \mathbf{M} = 1$, the transformation T preserves area.

1) b)

$\begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & \cosh x & \sinh x \\ 0 & \sinh x & \cosh x \end{pmatrix}$

So $O'(0,0)$, $A'(\cosh x\,, \sinh x)$ and $B'(\sinh x\,, \cosh x)$ are the images of O, A and B respectively.

$O'A'^2 = (\cosh x - 0)^2 + (\sinh x - 0)^2 = \cosh^2 x + \sinh^2 x$

$O'B'^2 = (\sinh x - 0)^2 + (\cosh x - 0)^2 = \sinh^2 x + \cosh^2 x$

$A'B'^2 = (\sinh x - \cosh x)^2 + (\cosh x - \sinh x)^2$

$= \sinh^2 x - 2\sinh x \cosh x + \cosh^2 x + \cosh^2 x - 2\sinh x \cosh x + \sinh^2 x$

$= 2\sinh^2 x - 4\sinh x \cosh x + 2\cosh^2 x$

The triangle is equilateral if $O'A' = O'B' = A'B'$

$\therefore 2\sinh^2 x - 4\sinh x \cosh x + 2\cosh^2 x = \cosh^2 x + \sinh^2 x$

$\Rightarrow \sinh^2 x + \cosh^2 x - 4\sinh x \cosh x = 0$

$\Rightarrow \cosh 2x - 2\sinh 2x = 0$

$\Rightarrow 1 - \frac{2\sinh 2x}{\cosh 2x} = 0$

$\Rightarrow 1 - 2\tanh 2x = 0$

$\Rightarrow \tanh 2x = \frac{1}{2}$

$\Rightarrow 2x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)$

$\Rightarrow x = \frac{1}{4}\ln 3$

2) a)

$y = 4\sinh x - 6\cosh x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4\cosh x - 6\sinh x$

Turning points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$4\cosh x - 6\sinh x = 0$

$\Rightarrow 4 - \frac{6\sinh x}{\cosh x} = 0$

$\Rightarrow 4 - 6\tanh x = 0$

$\Rightarrow \tanh x = \frac{2}{3}$

$\Rightarrow x = \frac{1}{2}\ln\left(\frac{1+\frac{2}{3}}{1-\frac{2}{3}}\right) = \frac{1}{2}\ln 5$ as required

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4\sinh x - 6\cosh x$

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=\frac{1}{2}\ln 5} = 4\sinh\left(\frac{1}{2}\ln 5\right) - 6\cosh\left(\frac{1}{2}\ln 5\right)$

$= 4 \times \frac{\mathrm{e}^{\frac{1}{2}\ln 5} - \mathrm{e}^{-\frac{1}{2}\ln 5}}{2} - 6 \times \frac{\mathrm{e}^{\frac{1}{2}\ln 5} + \mathrm{e}^{-\frac{1}{2}\ln 5}}{2}$

$= 2\left(\mathrm{e}^{\ln 5^{\frac{1}{2}}} - \mathrm{e}^{\ln 5^{-\frac{1}{2}}}\right) - 3\left(\mathrm{e}^{\ln 5^{\frac{1}{2}}} + \mathrm{e}^{\ln 5^{-\frac{1}{2}}}\right)$

$= 2\left(5^{\frac{1}{2}} - 5^{-\frac{1}{2}}\right) - 3\left(5^{\frac{1}{2}} + 5^{-\frac{1}{2}}\right)$

$= -5^{\frac{1}{2}} - 5 \times 5^{-\frac{1}{2}}$

$= -5^{\frac{1}{2}} - 5^{\frac{1}{2}}$

$= -2\sqrt{5} < 0$ so the turning point is a local maximum

2) b)

$V = \pi \int_0^{\frac{1}{2}\ln 5} (4\sinh x - 6\cosh x)^2 \, \mathrm{d}x$

$= \pi \int_0^{\frac{1}{2}\ln 5} (16\sinh^2 x - 48\sinh x \cosh x + 36\cosh^2 x) \, \mathrm{d}x$

$\cosh 2x \equiv 2\cosh^2 x - 1 \Rightarrow \cosh^2 x \equiv \frac{1}{2}(\cosh 2x + 1)$

$\cosh 2x \equiv 1 + 2\sinh^2 x \Rightarrow \sinh^2 x \equiv \frac{1}{2}(\cosh 2x - 1)$

$\sinh 2x \equiv 2\sinh x \cosh x$

$\therefore V = \pi \int_0^{\frac{1}{2}\ln 5} \left(16 \times \frac{1}{2}(\cosh 2x - 1) - 24\sinh 2x + 36 \times \frac{1}{2}(\cosh 2x + 1)\right) \mathrm{d}x$

$= \pi \int_0^{\frac{1}{2}\ln 5} (26\cosh 2x - 24\sinh 2x + 10) \, \mathrm{d}x$

$= \pi[13\sinh 2x - 12\cosh 2x + 10x]_0^{\frac{1}{2}\ln 5}$

$= \pi\left(\left(13\sinh\left(2 \times \frac{1}{2}\ln 5\right) - 12\cosh\left(2 \times \frac{1}{2}\ln 5\right) + 10 \times \frac{1}{2}\ln 5\right) - (13\sinh 0 - 12\cosh 0 + 10 \times 0)\right)$

$= \pi\left((13\sinh(\ln 5) - 12\cosh(\ln 5) + 5\ln 5) + 12\right)$

$= \pi\left(13 \times \frac{\mathrm{e}^{\ln 5} - \mathrm{e}^{-\ln 5}}{2} - 12 \times \frac{\mathrm{e}^{\ln 5} + \mathrm{e}^{-\ln 5}}{2} + 5\ln 5 + 12\right)$

$= \pi\left(13 \times \frac{\mathrm{e}^{\ln 5} - \mathrm{e}^{\ln 5^{-1}}}{2} - 12 \times \frac{\mathrm{e}^{\ln 5} + \mathrm{e}^{\ln 5^{-1}}}{2} + 5\ln 5 + 12\right)$

$= \pi\left(13 \times \frac{\mathrm{e}^{\ln 5} - \mathrm{e}^{\ln \frac{1}{5}}}{2} - 12 \times \frac{\mathrm{e}^{\ln 5} + \mathrm{e}^{\ln \frac{1}{5}}}{2} + 5\ln 5 + 12\right)$

$= \pi\left(13 \times \frac{5 - \frac{1}{5}}{2} - 12 \times \frac{5 + \frac{1}{5}}{2} + 5\ln 5 + 12\right)$

$= \pi\left(\frac{156}{5} - \frac{156}{5} + 5\ln 5 + 12\right)$

$= \pi(12 + 5\ln 5)$

So $p = 12$ and $q = 5$

3)

$r = 4\sqrt{\cosh 2\theta - \sinh 2\theta}$

$A = \frac{1}{2}\int_0^{\pi} r^2 \, \mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi} \left(4\sqrt{\cosh 2\theta - \sinh 2\theta}\right)^2 \mathrm{d}\theta$

$= \frac{1}{2}\int_0^{\pi} 16(\cosh 2\theta - \sinh 2\theta) \, \mathrm{d}\theta$

$= 8\int_0^{\pi} (\cosh 2\theta - \sinh 2\theta) \, \mathrm{d}\theta$

$= 8\left[\frac{1}{2}\sinh 2\theta - \frac{1}{2}\cosh 2\theta\right]_0^{\pi}$

$= 8\left(\left(\frac{1}{2}\sinh 2\pi - \frac{1}{2}\cosh 2\pi\right) - \left(\frac{1}{2}\sinh 0 - \frac{1}{2}\cosh 0\right)\right)$

$= 8\left(\left(\frac{1}{2} \times \frac{\mathrm{e}^{2\pi} - \mathrm{e}^{-2\pi}}{2} - \frac{1}{2} \times \frac{\mathrm{e}^{2\pi} + \mathrm{e}^{-2\pi}}{2}\right) - \left(0 - \frac{1}{2}\right)\right)$

$= 8\left(-\frac{1}{2}\mathrm{e}^{-2\pi} + \frac{1}{2}\right)$

$= 4 - 4\mathrm{e}^{-2\pi}$

4) a)

The point P is when $x = 0$:

$0 = 3\sinh t - \cosh t \Rightarrow 0 = \frac{3\sinh t}{\cosh t} - 1 \Rightarrow 0 = 3\tanh t - 1$

$\Rightarrow \tanh t = \frac{1}{3} \Rightarrow t = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$

$y|_{t=\frac{1}{2}\ln 2} = \sinh\left(\frac{1}{2}\ln 2\right) + 2\cosh\left(\frac{1}{2}\ln 2\right)$

$= \frac{\mathrm{e}^{\frac{1}{2}\ln 2} - \mathrm{e}^{-\frac{1}{2}\ln 2}}{2} + 2 \times \frac{\mathrm{e}^{\frac{1}{2}\ln 2} + \mathrm{e}^{-\frac{1}{2}\ln 2}}{2}$

$= \frac{1}{2}\left(\mathrm{e}^{\ln 2^{\frac{1}{2}}} - \mathrm{e}^{\ln 2^{-\frac{1}{2}}}\right) + \mathrm{e}^{\ln 2^{\frac{1}{2}}} + \mathrm{e}^{\ln 2^{-\frac{1}{2}}}$

$= \frac{1}{2}\mathrm{e}^{\ln\sqrt{2}} - \frac{1}{2}\mathrm{e}^{\ln\frac{\sqrt{2}}{2}} + \mathrm{e}^{\ln\sqrt{2}} + \mathrm{e}^{\ln\frac{\sqrt{2}}{2}}$

$= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{4} + \sqrt{2} + \frac{\sqrt{2}}{2}$

$= \frac{7\sqrt{2}}{4}$

So P has cooridnates $\left(0, \frac{7\sqrt{2}}{4}\right)$

4) b)

$x|_{t=0} = 3\sinh 0 - \cosh 0 = -1$

$y|_{t=0} = \sinh 0 + 2\cosh 0 = 2$

So Q has coordinates $(-1, 2)$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 3\cosh t - \sinh t$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \cosh t + 2\sinh t$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}} = \frac{\cosh t + 2\sinh t}{3\cosh t - \sinh t}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{t=0} = \frac{\cosh 0 + 2\sinh 0}{3\cosh 0 - \sinh 0} = \frac{1+0}{3-0} = \frac{1}{3}$

So the tangent has equation:

$y - 2 = \frac{1}{3}(x + 1) \Rightarrow 3y - 6 = x + 1$

$\Rightarrow -x + 3y = 7$

5)

$x = 5\sinh t - 8\cosh t$

$y = 2\cosh t - 3\sinh t$

$\therefore x + y = 2\sinh t - 6\cosh t$

Maximum value of $x + y$ occurs when $\frac{\mathrm{d}}{\mathrm{d}t}(2\sinh t - 6\cosh t) = 0$

$\frac{\mathrm{d}}{\mathrm{d}t}(2\sinh t - 6\cosh t) = 2\cosh t - 6\sinh t$

$2\cosh t - 6\sinh t = 0 \Rightarrow 1 - 3\tanh t = 0 \Rightarrow \tanh t = \frac{1}{3}$

$\Rightarrow t = \operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$

$\max(x + y) = 2\sinh\left(\frac{1}{2}\ln 2\right) - 6\cosh\left(\frac{1}{2}\ln 2\right)$

$= 2 \times \frac{\mathrm{e}^{\frac{1}{2}\ln 2} - \mathrm{e}^{-\frac{1}{2}\ln 2}}{2} - 6 \times \frac{\mathrm{e}^{\frac{1}{2}\ln 2} + \mathrm{e}^{-\frac{1}{2}\ln 2}}{2}$

$= \mathrm{e}^{\frac{1}{2}\ln 2} - \mathrm{e}^{-\frac{1}{2}\ln 2} - 3\left(\mathrm{e}^{\frac{1}{2}\ln 2} + \mathrm{e}^{-\frac{1}{2}\ln 2}\right)$

$= -2\mathrm{e}^{\frac{1}{2}\ln 2} - 4\mathrm{e}^{-\frac{1}{2}\ln 2}$

$= -2\mathrm{e}^{\ln 2^{\frac{1}{2}}} - 4\mathrm{e}^{\ln 2^{-\frac{1}{2}}}$

$= -2\mathrm{e}^{\ln(\sqrt{2})} - 4\mathrm{e}^{\ln\left(\frac{\sqrt{2}}{2}\right)}$

$= -2 \times \sqrt{2} - 4 \times \frac{\sqrt{2}}{2}$

$= -4\sqrt{2}$

$\therefore x + y = -4\sqrt{2}$

6) a)

$x^2 + 3y^2 = (2\sinh 2t - \cosh 2t)^2 + 3(\sinh 2t)^2$

$= 4\sinh^2 2t - 4\sinh 2t\cosh 2t + \cosh^2 2t + 3\sinh^2 2t$

$= 7\sinh^2 2t - 4\sinh 2t\cosh 2t + \cosh^2 2t$

$= 7\sinh^2 2t - 4\sinh 2t\cosh 2t + (1 + \sinh^2 2t)$

$= 8\sinh^2 2t - 4\sinh 2t\cosh 2t + 1$

$= 4(2\sinh 2t - \cosh 2t)\sinh 2t + 1$

$= 4xy + 1$ as required

6) b)

$A = \int_0^{\ln 2} y \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$

$y = \sinh 2t$

$x = 2\sinh 2t - \cosh 2t$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 4\cosh 2t - 2\sinh 2t$

$\therefore A = \int_0^{\ln 2} \sinh 2t \times (4\cosh 2t - 2\sinh 2t)\,\mathrm{d}t$

$= \int_0^{\ln 2} (4\sinh 2t \cosh 2t - 2\sinh^2 2t)\,\mathrm{d}t$

Note that $\sinh 2A \equiv 2\sinh A \cosh A$ and $\cosh 2A \equiv 1 + 2\sinh^2 A$

So $\sinh 4t \equiv 2\sinh 2t \cosh 2t$ and $\cosh 4t = 1 + 2\sinh^2 2t$

$\therefore A = \int_0^{\ln 2} (2\sinh 4t - \cosh 4t + 1)\,\mathrm{d}t$

$= \left[\frac{1}{2}\cosh 4t - \frac{1}{4}\sinh 4t + t\right]_0^{\ln 2}$

$= \left(\frac{1}{2} \times \frac{\mathrm{e}^{4\ln 2} + \mathrm{e}^{-4\ln 2}}{2} - \frac{1}{4} \times \frac{\mathrm{e}^{4\ln 2} - \mathrm{e}^{-4\ln 2}}{2} + \ln 2\right) - \left(\frac{1}{2} - 0 + 0\right)$

$= \left(\frac{1}{4}\left(\mathrm{e}^{\ln 2^4} + \mathrm{e}^{\ln 2^{-4}}\right) - \frac{1}{8}\left(\mathrm{e}^{\ln 2^4} - \mathrm{e}^{\ln 2^{-4}}\right) + \ln 2\right) - \frac{1}{2}$

$= \left(\frac{1}{4}\left(\mathrm{e}^{\ln 2^4} + \mathrm{e}^{\ln 2^{-4}}\right) - \frac{1}{8}\left(\mathrm{e}^{\ln 2^4} - \mathrm{e}^{\ln 2^{-4}}\right) + \ln 2\right) - \frac{1}{2}$

$= \frac{1}{4}\left(\mathrm{e}^{\ln 16} + \mathrm{e}^{\ln \frac{1}{16}}\right) - \frac{1}{8}\left(\mathrm{e}^{\ln 16} - \mathrm{e}^{\ln \frac{1}{16}}\right) + \ln 2 - \frac{1}{2}$

$= \frac{1}{4}\left(16 + \frac{1}{16}\right) - \frac{1}{8}\left(16 - \frac{1}{16}\right) + \ln 2 - \frac{1}{2}$

$= \frac{257}{64} - \frac{255}{128} + \ln 2 - \frac{1}{2}$

$= \frac{195}{128} + \ln 2$

7) a)

$\cosh(\mathrm{i}z) = \frac{\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}}{2} = \frac{(\cos z + \mathrm{i}\sin z) + (\cos z - \mathrm{i}\sin z)}{2} = \frac{2\cos z}{2} = \cos z$

7) b)

$\cos z = 4$

$\Rightarrow \cosh(\mathrm{i}z) = 4$

$\Rightarrow \mathrm{i}z = \pm \operatorname{arcosh} 4$

$\Rightarrow \mathrm{i}z = \pm \ln\left(4 + \sqrt{4^2 - 1}\right)$

$\Rightarrow \mathrm{i}z = \pm \ln\left(4 + \sqrt{15}\right)$

$\Rightarrow z = \pm \frac{\ln\left(4+\sqrt{15}\right)}{\mathrm{i}}$

$\Rightarrow z = \pm \mathrm{i} \ln\left(4 + \sqrt{15}\right)$

Or $z = \mathrm{i} \ln\left(4 \pm \sqrt{15}\right)$

8) a)

$3 \sinh x + 5 \cosh x$

$\equiv 3 \times \frac{\mathrm{e}^x - \mathrm{e}^{-x}}{2} + 5 \times \frac{\mathrm{e}^x + \mathrm{e}^{-x}}{2}$

$\equiv \frac{3}{2}(\mathrm{e}^x - \mathrm{e}^{-x}) + \frac{5}{2}(\mathrm{e}^x + \mathrm{e}^{-x})$

$\equiv \frac{3}{2}\mathrm{e}^x - \frac{3}{2}\mathrm{e}^{-x} + \frac{5}{2}\mathrm{e}^x + \frac{5}{2}\mathrm{e}^{-x}$

$\equiv 4\mathrm{e}^x + \mathrm{e}^{-x}$

$\equiv \mathrm{e}^{-x}(1 + 4\mathrm{e}^{2x})$

8) b)

$$\int_{\ln\frac{\sqrt{3}}{6}}^{\ln\frac{1}{2}} \frac{1}{3\sinh x+5\cosh x}\,\mathrm{d}x$$

$$= \int_{\ln\frac{\sqrt{3}}{6}}^{\ln\frac{1}{2}} \frac{1}{\mathrm{e}^{-x}(1+4\mathrm{e}^{2x})}\,\mathrm{d}x$$

$$= \int_{\ln\frac{\sqrt{3}}{6}}^{\ln\frac{1}{2}} \frac{\mathrm{e}^{x}}{1+4\mathrm{e}^{2x}}\,\mathrm{d}x$$

Notice that the denominator is $1+u^2$ if $u = 2\mathrm{e}^{x}$

$u = 2\mathrm{e}^{x}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2\mathrm{e}^{x}$

$\Rightarrow \mathrm{d}x = \frac{1}{2\mathrm{e}^{x}}\,\mathrm{d}u$

Change Limits:

$x = \ln\frac{1}{2} \Rightarrow u = 2\mathrm{e}^{\ln\frac{1}{2}} = 1$

$x = \ln\frac{\sqrt{3}}{6} \Rightarrow u = 2\mathrm{e}^{\ln\frac{\sqrt{3}}{6}} = \frac{\sqrt{3}}{3}$

$$\therefore \int_{\ln\frac{\sqrt{3}}{6}}^{\ln\frac{1}{2}} \frac{\mathrm{e}^{x}}{1+4\mathrm{e}^{2x}}\,\mathrm{d}x = \int_{\frac{\sqrt{3}}{3}}^{1} \frac{\mathrm{e}^{x}}{1+u^2} \times \frac{1}{2\mathrm{e}^{x}}\,\mathrm{d}u$$

$$= \frac{1}{2}\int_{\frac{\sqrt{3}}{3}}^{1} \frac{1}{1+u^2}\,\mathrm{d}u$$

$$= \frac{1}{2}\left[\arctan u\right]_{\frac{\sqrt{3}}{3}}^{1}$$

$$= \frac{1}{2}\left(\arctan 1 - \arctan\frac{\sqrt{3}}{3}\right)$$

$$= \frac{1}{2}\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$$

$$= \frac{\pi}{24}$$

9) a)

$$\mathrm{f}(0) = \operatorname{artanh}\left(\frac{2\times0-1}{2-2\times0}\right)$$
$$= \operatorname{artanh}\left(-\frac{1}{2}\right)$$
$$= \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{2}\right)}{1-\left(-\frac{1}{2}\right)}\right)$$
$$= \frac{1}{2}\ln\left(\frac{1}{3}\right)$$
$$= \ln\left(\left(\frac{1}{3}\right)^{\frac{1}{2}}\right)$$
$$= \ln\left(\frac{1}{\sqrt{3}}\right)$$
$$= \ln\left(\frac{\sqrt{3}}{3}\right)$$

9) b)

$$\mathrm{f}(x) = \operatorname{artanh}\left(\frac{2x-1}{2-2x}\right)$$
$$\mathrm{f}'(x) = \frac{(2-2x)\times2-(2x-1)\times(-2)}{(2-2x)^2} \times \frac{1}{1-\left(\frac{2x-1}{2-2x}\right)^2}$$
$$= \frac{2}{(2-2x)^2} \times \frac{1}{1-\frac{(2x-1)^2}{(2-2x)^2}}$$
$$= \frac{2}{(2-2x)^2} \times \frac{(2-2x)^2}{(2-2x)^2-(2x-1)^2}$$
$$= \frac{2}{1} \times \frac{1}{4-8x+4x^2-(4x^2-4x+1)}$$
$$= \frac{2}{3-4x}$$

9) c)

$$\mathrm{f}''(x) = \frac{(3-4x)\times0-2\times(-4)}{(3-4x)^2}$$
$$= \frac{8}{(3-4x)^2}$$

9) d)

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots$$

$$f(0) = \ln\left(\frac{\sqrt{3}}{3}\right)$$

$$f'(0) = \frac{2}{3-4\times 0} = \frac{2}{3}$$

$$f''(0) = \frac{8}{(3-4\times 0)^2} = \frac{8}{9}$$

$$\therefore f(x) = \ln\left(\frac{\sqrt{3}}{3}\right) + \frac{2}{3}x + \frac{\frac{8}{9}}{2!}x^2 + \cdots$$

$$\approx \ln\left(\frac{\sqrt{3}}{3}\right) + \frac{2}{3}x + \frac{4}{9}x^2$$

10)

$$\mathrm{f}(x) = \sinh \mathrm{i}x = \frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2}$$
$$= \frac{(\cos x + \mathrm{i}\sin x) - (\cos x - \mathrm{i}\sin x)}{2}$$
$$= \frac{2\mathrm{i}\sin x}{2}$$
$$= \mathrm{i}\sin x$$
$$y = \big(\mathrm{f}(x)\big)^4 = (\mathrm{i}\sin x)^4 = \sin^4 x$$

Mean Value:

$$= \frac{1}{2\pi - 0}\int_0^{2\pi} \sin^4 x \,\mathrm{d}x$$

Now:

$$(\cos x + \mathrm{i}\sin x)^4 = \cos 4x + \mathrm{i}\sin 4x$$
$$\sin x = \frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2\mathrm{i}}$$
$$\Rightarrow \sin^4 x = \left(\frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2\mathrm{i}}\right)^4$$
$$= \frac{\left(\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}\right)^4}{16}$$
$$= \frac{\mathrm{e}^{4\mathrm{i}x} - 4\mathrm{e}^{2\mathrm{i}x} + 6 - 4\mathrm{e}^{-2\mathrm{i}x} + \mathrm{e}^{-4\mathrm{i}x}}{16}$$
$$= \frac{\mathrm{e}^{4\mathrm{i}x} + \mathrm{e}^{-4\mathrm{i}x} - 4\left(\mathrm{e}^{2\mathrm{i}x} + 4\mathrm{e}^{-2\mathrm{i}x}\right) + 6}{16}$$

Meanwhile, $\cos\theta = \frac{\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}}{2} \Rightarrow \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$

$$\therefore \sin^4 x = \frac{2\cos 4x - 8\cos 2x + 6}{16} = \frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}$$

So the Mean Value:

$$= \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}\right)\mathrm{d}x$$
$$= \frac{1}{2\pi}\left[\frac{1}{32}\sin 4x - \frac{1}{4}\sin 2x + \frac{3}{8}x\right]_0^{2\pi}$$
$$= \frac{1}{2\pi}\left(\left(0 - 0 + \frac{3\pi}{4}\right) - (0 - 0 + 0)\right) = \frac{1}{2\pi} \times \frac{3\pi}{4} = \frac{3}{8}$$

Alternatively:

$$\mathrm{f}(x) = \sinh \mathrm{i}x = \frac{\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x}}{2}$$
$$= \frac{(\cos x + \mathrm{i}\sin x) - (\cos x - \mathrm{i}\sin x)}{2}$$
$$= \frac{2\mathrm{i}\sin x}{2}$$
$$= \mathrm{i}\sin x$$
$$y = \big(\mathrm{f}(x)\big)^4 = (\mathrm{i}\sin x)^4 = \sin^4 x$$

Mean Value:

$= \frac{1}{2\pi - 0}\int_0^{2\pi} \sin^4 x \, dx$

$\sin^4 x = (\sin^2 x)^2 = \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right)^2$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}\cos^2 2x$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos 4x\right)$

$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{8} + \frac{1}{8}\cos 4x$

$= \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$

So the Mean Value:

$= \frac{1}{2\pi}\int_0^{2\pi} \left(\frac{1}{8}\cos 4x - \frac{1}{2}\cos 2x + \frac{3}{8}\right) dx$

$= \frac{1}{2\pi}\left[\frac{1}{32}\sin 4x - \frac{1}{4}\sin 2x + \frac{3}{8}x\right]_0^{2\pi}$

$= \frac{1}{2\pi}\left(\left(0 - 0 + \frac{3\pi}{4}\right) - (0 - 0 + 0)\right) = \frac{1}{2\pi} \times \frac{3\pi}{4} = \frac{3}{8}$

11) a)

$\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B$

$\cosh\big((p+1)x\big) = \cosh(px + x) = \cosh px \cosh x + \sinh px \sinh x$

$\cosh\big((p-1)x\big) = \cosh(px - x) = \cosh px \cosh(-x) + \sinh px \sinh(-x)$

$= \cosh px \cosh x - \sinh px \sinh x$

11) b)

$\cosh\big((p+1)x\big) - \cosh\big((p-1)x\big)$

$= (\cosh px \cosh x + \sinh px \sinh x) - (\cosh px \cosh x - \sinh px \sinh x)$

$= 2 \sinh px \sinh x$

So $\cosh\big((p+1)x\big) - \cosh\big((p-1)x\big) = 2 \sinh px \sinh x$

$\Rightarrow \sinh px = \frac{\cosh\big((p+1)x\big) - \cosh\big((p-1)x\big)}{2 \sinh x}$

11) c)

$S_n = \sinh x + \sinh 2x + \cdots + \sinh nx$

$= \sum_{k=1}^{n} \frac{\cosh((k+1)x) - \cosh((k-1)x)}{2\sinh x}$

$k = 1$: $\frac{\cancel{\cosh 2x} - \cosh 0}{2\sinh x}$

$k = 2$: $\frac{\cancel{\cosh 3x} - \cosh x}{2\sinh x}$

$k = 3$: $\frac{\cancel{\cosh 4x} - \cancel{\cosh 2x}}{2\sinh x}$

…

$k = n - 2$: $\frac{\cancel{\cosh((n-1)x)} - \cancel{\cosh((n-3)x)}}{2\sinh x}$

$k = n - 1$: $\frac{\cosh nx - \cancel{\cosh((n-2)x)}}{2\sinh x}$

$k = n$: $\frac{\cosh((n+1)x) - \cancel{\cosh((n-1)x)}}{2\sinh x}$

$\therefore S_n = \frac{\cosh nx}{2\sinh x} + \frac{\cosh((n+1)x)}{2\sinh x} - \frac{\cosh 0}{2\sinh x} - \frac{\cosh x}{2\sinh x}$

$= \frac{\cosh nx + \cosh((n+1)x) - \cosh x - 1}{2\sinh x}$

12)

$\frac{dy}{dx} + y\coth x = \cosh^2 x$

Integrating Factor: $e^{\int \coth x dx} = e^{\ln|\sinh x|} = \sinh x$

$\Rightarrow \sinh x \frac{dy}{dx} + y\sinh x \coth x = \sinh x \cosh^2 x$

$\Rightarrow \sinh x \frac{dy}{dx} + y\cosh x = \sinh x \cosh^2 x$

$\Rightarrow \frac{d}{dx}(y\sinh x) = \sinh x \cosh^2 x$

$\Rightarrow y\sinh x = \frac{1}{3}\cosh^3 x + c$

$y = 2$ when $x = \ln 2$:

$2\sinh(\ln 2) = \frac{1}{3}\cosh^3(\ln 2) + c$

$\Rightarrow 2 \times \frac{e^{\ln 2} - e^{-\ln 2}}{2} = \frac{1}{3} \times \left(\frac{e^{\ln 2} + e^{-\ln 2}}{2}\right)^3 + c$

$\Rightarrow \frac{3}{2} = \frac{125}{192} + c$

$\Rightarrow c = \frac{163}{192}$

$\therefore y\sinh x = \frac{1}{3}\cosh^3 x + \frac{163}{192}$

$\Rightarrow y = \frac{\cosh^3 x}{3\sinh x} + \frac{163}{192\sinh x}$

13)

$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tanh x = \sinh 2x$

Integrating Factor: $\mathrm{e}^{\int -\tanh x \mathrm{d}x} = \mathrm{e}^{-\ln|\cosh x|} = \mathrm{e}^{\ln\left|\frac{1}{\cosh x}\right|} = \mathrm{e}^{\ln|\operatorname{sech} x|}$

$\Rightarrow \operatorname{sech} x \frac{\mathrm{d}y}{\mathrm{d}x} - y\operatorname{sech} x \tanh x = \operatorname{sech} x \sinh 2x$

$\Rightarrow \operatorname{sech} x \frac{\mathrm{d}y}{\mathrm{d}x} - y\operatorname{sech} x \tanh x = 2\sinh x$

$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}(y\operatorname{sech} x) = 2\sinh x$

$\Rightarrow y\operatorname{sech} x = 2\cosh x + c$

$y = 5$ when $x = 0$:

$5\operatorname{sech} 0 = 2\cosh 0 + c$

$\Rightarrow 5 = 2 + c$

$\Rightarrow c = 3$

$\therefore y\operatorname{sech} x = 2\cosh x + 3$

$\Rightarrow y = 2\cosh^2 x + 3\cosh x$